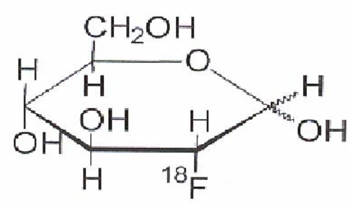
FDG PUTRA INJECTION

Description
FDG Putra Injection is a radio labeled analog of glucose which is used for diagnostic purposes in conjunction with Positron Emission Tomography (PET). It is administered by intravenous route.

Active Ingredient
The active ingredient of FDG Putra Injection is 2-deoxy-2-[^{18}F] fluoro-D-glucose, abbreviated [^{18}F] FDG, has a molecular weight of $C_6H_{11}{}^{18}FO_5$ with a molecular weight of 181.26 daltons, and has the following chemical structure:



FDG Putra Injection is supplied as a ready to use sterile, pyrogen free, clear and colourless solution. The pH of the solution is between 4.5-8.5. This solution is packed in a 10 ml sterile glass vial. In each mL of FDG Putra Injection contains 370-9250 MBq (10-250 mCi).

Physical Characteristic
Fluorine F-18 decays by positron ($ß^+$) emission and has a half-life of 109.7 minutes. The principal photons useful for diagnostic imaging are the 511 keV gamma photons, resulting from the interaction of the emitted positron with an electron (Table 1).

Table 1.Principal Emission Data for Fluorine F 18

Radiation/Emission	**% per Disintegration**	**Mean Energy**
Positron (β^+)	96.73	249.8 keV
Gamma (±)*	193.46	511.0 keV

*Produced by positron annihilation
From: Kocher, D.C. “Radioactive Decay Tables” DOE/TIC-11062,89 (1981)

Table Decay Factors
For use in correcting for physical decay of this radionuclide, the fractions remaining at selected intervals after calibration are shown in Table 2.

Table 2. Physical Decay Chart for Fluorine F 18

Minutes	Fraction Remaining
0*	1.00
15	0.909
30	0.826
60	0.683
110	0.500
220	0.250
440	0.060

* Calibration time

Pharmacodynamics
FDG Putra Injection is rapidly distributed to all organs of the body after intravenous administration. After background clearance of FDG Putra Injection, optimal PET imaging is generally achieved between 30 to 40 minutes after administration.

In cancer, the cells are generally characterized by enhanced glucose metabolism partially due to (1) an increase in activity of glucose transporters, (2) an increased rate of phosphorylation activity, (3) a reduction of phosphatase activity or, (4) a dynamic alteration in the balance among all these processes. However, glucose metabolism of cancer as reflected by FDG Putra Injection accumulation shows variability. It depends on tumor type, stage and location whereby the FDG Putra Injection accumulation may be increased, normal or decreased. This goes the same for inflammatory cells.

In the heart, under normal aerobic conditions, the myocardium meets the bulk of its energy requirements by oxidizing free fatty acids. Most of the exogenous glucose taken up by the myocyte is converted into glycogen. However, under ischemic conditions, the oxidation of free fatty acids decreases, exogenous glucose becomes the preferred myocardial substrate, glycolysis is stimulated, and glucose taken up by the myocyte is metabolized immediately instead of being converted into glycogen. Under these conditions, phosphorylated FDG Putra Injection accumulated in the myocyte and can be detected with PET imaging.

Pharmacokinetics

Absorption

Due to intravenous administration, the bioavailability of FDG Putra Injection is 100%.

Distribution

FDG Putra Injection is an analog of glucose and is used as a tracer of glucose metabolism. Therefore, the distribution of FDG Putra Injection is not limited to malignant tissue. Myocardial uptake is highly variable ranging from very intense to nil. Salivary gland and tonsillar uptakes are commonly seen and are somewhat variable. The lungs typically show very low uptake with somewhat higher uptake in the mediastinum. The liver shows modest uptake (usually slightly higher than mediastinum) with somewhat less intense uptake seen in the spleen. Uptake in the gastrointestinal (GI) tract (esophagus, stomach, colon) is highly variable. Since FDG Putra Injection is excreted by the kidneys, there is frequently intense FDG activity in the renal collecting system and urinary bladder, with less activity in the renal cortex. Muscle shows low uptake at rest. Mild activity is seen in hematopoietic bone marrow but no bone uptake is present. Thyroid uptake is usually minimal and nonspecific. Breast uptake is variable with mild to moderate. Glandular uptake commonly seen in premenopausal women and with estrogen therapy. Relatively intense uptake is seen in lactating breasts, but the uptake is largely glandular with little activity in breast milk. Generally, low breast uptake is present in the postmenopausal state.

Plasma Protein Binding

The extend of binding of FDG Putra Injection to plasma protein is not known.

Metabolism

FDG Putra Injection is transported into cells and phosphorylated to [^{18}F]-FDG-6-phosphate at a rate proportional to the rate of glucose utilization within that tissue. [^{18}F]-FDG-6-phosphate presumably is metabolized to 2-deoxy-2-[^{18}F] fluoro-6-phospho-D-mannose ([^{18}F] FDM-6-phosphate).

FDG Putra Injection may contain several impurities (e.g., 2-deoxy-2-chloro-D-glucose (CIDG). Biodistribution and metabolism of CIDG are presumed to be similar to FDG Putra Injection and would be expected to result in intracellular formation of 2-deoxy-2-chloro-phospho-D-glucose (CIDG-6-phosphate) and 2-deoxy-2-chloro-6-phospho-D-mannose (CIDM-6-phosphate). The phosphorylated deoxyglucose compounds are dephosphorylated and the resulting compounds (FDG, FDM, CIDG and CIDM) presumably leave cells by passive diffusion.

FDG Putra Injection and related compound are cleared from non-cardiac tissues within 3 to 24 hours after administration. Clearance from the cardiac tissue may require more than 96 hours.

FDG Putra Injection that is not involved in glucose metabolism in any tissue is then excreted in the urine.

Excretion

FDG Putra Injection is cleared from most tissues within 24 hours, and can be eliminated from the body unchanged in the urine. Three elimination phases have been identified in the reviewed literature. Within 33 minutes, a mean of 3.9% of the administered radioactive dose was measured in the urine. The amount of radiation exposure of the urinary bladder at two hours post administration suggests that 20.6% (mean) of the radioactive dose was present in the bladder.

Indication

FDG Putra Injection is indicated in positron emission tomography (PET) imaging for assessment of abnormal glucose metabolism to assist in the evaluation of malignancy in patients with known or suspected abnormalities found by other testing modalities, or in patients with an existing diagnosis of cancer.

FDG Putra Injection is indicated in positron emission tomography (PET) imaging in patients with coronary artery disease and left ventricular dysfunction, when used together with myocardial perfusion imaging, for the identification of left ventricular myocardium with residual glucose metabolism and reversible loss of systolic function.

FDG Putra Injection is indicated in positron emission tomography (PET) imaging in patients for the identification of regions of abnormal glucose metabolism associated with foci of epileptic seizure.

Recommended Dosage

The recommended dose is 6 Mbq/kg (0.162 mCi/kg)
Example, for an adult with a body weight of 70 kg, the recommended dose is 420 Mbq (11 mCi)

Mode of Administration

In general, FDG Putra Injection should be administered after patients have fasted for 4-6 hours. For cardiac use, FDG Putra Injection may be administered either to patients who have fasted or to patients who have received a glucose load (See Patient Preparation section).

Patient Preparation

Blood glucose levels should be stabilized before FDG Putra Injection is administered. In non-diabetic patients this may be accomplished by fasting 4-6 hours before FDG Putra Injection. Diabetic patients may need stabilization of blood glucose on the day preceding and on the day of administration of FDG Putra Injection. The patient must be advice on not to perform any strenuous exercise on the day before scan and also on the day of scan.

a) Medications

- The patient should stop all iron-containing medications at least one day before scan
- Avoid any diabetic medications on the morning prior to scan.

b) Diet

- On the day of scanning, the patient shall not eat or drink (except plain water)
- Please avoid the following:
    - red meat
    - brown bread
    - yellow cheese
    - cereals
    - fruits
    - vegetables

Imaging

Optimally, it is recommended that positron emission tomography (PET) imaging be initiated within 40 minutes of administration of FDG Putra Injection.

Contraindication

It is contraindicated in pregnancy and lactating woman.

Warning and Precaution

Patient should avoid extreme activity and the blood glucose levels should be stabilized before FDG Putra Injection is administered (See Patient Preparation section). The patient should be rest and keep isolated from anybody for at least 1 hour and after injection.

Interaction with Other Medications

All iron containing medications and diabetic medications (See Patient Preparation section)

Side Effects

Not known

Overdosage

Overdoses of FDG Putra Injection have not been reported. See Radiation Dosimetry section for related information

Radiation Dosimetry

The estimated human absorbed radiation doses (rem/mCi) to a 1-year old (9.8 kg), 5-year old (19 kg), 10-year old (32 kg), 15-year old (57 kg), and adult (70 kg) from intravenous administration of FDG Putra Injection are shown in Table 3. These estimates were calculated based on human data and using the data published by the International Commission on Radiological Protection for Fludeoxyglucose F-18. The dosimetry data obtained and presented in this table show that there are slight variations in absorbed radiation dose for various organs in each of the age groups. These dissimilarities in absorbed radiation dose are understood to be due to developmental age variations (e.g., organ size, location, and overall metabolic rate for each age group). The identified critical organs (in descending order) across all age groups evaluated (i.e., newborn, 1, 5, 10, 15 year(s) and adults) are the urinary bladder, heart, pancreas, spleen, and lungs. The absolute values for absorbed radiation in each of these organs vary in each of the age groups.

Table 3. Estimated Absorbed Radiation Doses (rem/mCi) After Intravenous Aministration of Fludeoxyglucose F-18

Organ	Newborn (3.4 kg)	1-year old (9.8 kg)	5-year old (19 kg)	10-year old (32 kg)	15-year old (32 kg)	Adult (70 kg)
Bladder Wall	4.3	1.7	0.93	0.60	0.40	0.32
Heart Wall	2.4	1.2	0.70	0.44	0.29	0.22
Pancreas	2.2	0.68	0.33	0.25	0.13	0.096
Spleen	2.2	0.84	0.46	0.29	0.19	0.14
Lungs	0.96	0.38	0.20	0.13	0.092	0.064
Kidneys	0.81	0.34	0.19	0.13	0.089	0.074
Ovaries	0.80	0.80	0.19	0.11	0.058	0.053
Uterus	0.79	0.35	0.19	0.12	0.076	0.062
LLI wall	0.69	0.28	0.15	0.097	0.060	0.051
Liver	0.69	0.31	0.17	0.11	0.076	0.025
Gallbladder wall	0.69	0.26	0.14	0.093	0.059	0.049
Sm Intestine	0.68	0.29	0.15	0.096	0.060	0.047
ULI wall	0.67	0.27	0.15	0.090	0.057	0.046
Stomach wall	0.65	0.27	0.14	0.089	0.057	0.047
Adrenals	0.65	0.28	0.15	0.095	0.061	0.048
Testes	0.64	0.27	0.14	0.085	0.052	0.041
Red marrow	0.62	0.26	0.14	0.089	0.057	0.047
Thymus	0.61	0.26	0.14	0.086	0.056	0.044
Thyroid	0.61	0.26	0.13	0.080	0.049	0.039
Muscle	0.58	0.25	0.13	0.078	0.0469	0.039
Bone surfaces	0.57	0.24	0.12	0.079	0.052	0.041
Breast	0.54	0.22	0.11	0.068	0.043	0.034
Skin	0.49	0.20	0.10	0.060	0.037	0.030
Brain	0.29	0.13	0.13	0.078	0.072	0.070
Other tissues	0.59	0.25	0.25	0.083	0.052	0.042

Incompatibilities
Not known

Symptoms and Treatment of overdose
Overdoses of FDG Putra Injection have not been reported

Storage Condition
FDG Putra Injection should be store below 25^0C

Shelf Life
The shelf life for FDG Putra Injection is 12 hours after calibration

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